Engineering an Interpretable Temporal Event Pipeline with Explicit Channel Availability: A Combat-Sports Case Study

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Abstract

Multi-source temporal event detection for engineering decision support requires pipelines that remain interpretable when individual evidence channels are noisy or intermittently unavailable. This study evaluates a modular, availability-aware pipeline that separates held-fixed perception features, soft channel confidences c, binary availability masks α, frame-wise aggregation, optional interaction-rule boosts, score banding, and temporal consolidation. Unavailable channels (α=0) are treated as distinct from zero-valued evidence (c=0, α=1). On BoxingVI (n=10) with punch/impact *proxy* labels and a *combined timeline* in which only the risk component is re-aggregated, equal and weighted aggregation yield nearly identical pooled micro-F1 (0.502), whereas max aggregation collapses to 0.178. Configured interaction rules raise macro-F1 from 0.444 to 0.567 (paired permutation p=0.048), a protocol-limited effect. Under synthetic channel dropout at p=0.5, explicit availability masking (0.509) is less degraded than naive zero-valued evidence (0.471); natural availability on these matrices is $\alpha\equiv$1, and pooled metrics are dominated by one stem. The study is an engineering evaluation of pipeline levers: not a universal solution to partial observability, not a deployment-ready safety system, and not an external state-of-the-art detector comparison.

Keywords: interpretable AI, temporal event detection, channel availability, decision support, reproducible pipeline, combat-sports case study,

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1. Introduction

Engineering AI systems that turn heterogeneous evidence into temporal intervals face a practical design problem before accuracy debates begin. In review workflows and offline analysis pipelines, perception stages emit soft channel confidences at different reliabilities; some channels may drop out; and downstream users need intervals that can be inspected channel by channel [1–3]. When detectors, aggregators and consolidators are stacked without an explicit contract, comparisons become unfair, missing channels are silently treated as zero-valued evidence, and negative ablation outcomes are discarded instead of informing design [4, 5].

This manuscript targets *Engineering Applications of Artificial Intelligence*, a venue that regularly publishes applied AI system evaluations and engineering analyses of learning pipelines [6]. The systems problem is studied on BoxingVI [7] through a controlled evaluation of an engineered artefact: modular stages, declared interfaces, controlled ablations and reported failure modes under a fixed protocol. Perception features are frozen; a common frame-level matrix of channel confidences c and availability masks α is constructed; and only interpretable aggregation and interaction settings are varied under one consolidator and one matcher. What varies is aggregation scheme, interaction-rule enablement and synthetic availability encoding. What is held fixed includes perception features, channel weights, thresholds, rules, consolidator parameters, matcher settings and the strike component of the combined timeline. Outside scope are clinical or officiating deployment, operator-in-the-loop outcome studies, end-to-end deep detectors trained for BoxingVI leaderboards, and claims of solving natural partial observability in general. Null and negative findings (equal≈weighted; brittle max; synthetic-only missingness) are treated as engineering evidence because they constrain where tuning effort should and should not be spent [4, 8].

The scientific object is therefore a *reproducible pipeline contract* and the empirical behaviour of its levers under disclosed limits. The contributions are: (i) a reproducible multi-source pipeline separating features, (c, α), aggregation, interaction rules, banding and consolidation [1, 4]; (ii) a same-input comparison of equal, weighted and max aggregation (max as stress test) under a frozen consolidator and matcher [9]; (iii) an interaction-rule ablation with firing-incidence counts (n=10; borderline paired evidence) [3, 10]; and (iv) a controlled *synthetic* missingness contrast of explicit α=0 availability masking versus naive zero-valued evidence [11–13]. All numerical claims below are taken from the canonical checkpoint `run_20260730_005150`.

2. Related work

Deep temporal action localisation and temporal proposal generation provide strong learned detectors when labels and end-to-end training are available [9, 14–16]. In sports video, action-spotting protocols such as SoccerNet emphasise sparse temporal events under shared evaluation discipline [17], while human action recognition surveys place combat and team-sport analytics within broader video understanding [18, 19]. BoxingVI supplies multimodal boxing footage with temporal annotations suitable for interval studies and is adopted here as an engineering case study with punch/impact *proxy* labels [7]. These lines establish what high-capacity temporal models can achieve when perception and detection are trained jointly. They leave aggregation, availability handling and rule interactions rarely isolated as controlled engineering variables. The present study does not propose a new localisation architecture or compete for BoxingVI detection leaderboards; it uses a held-fixed perception stage so that aggregation and availability contracts can be compared under one matcher, without validating clinical safety monitoring or automated officiating.

Decision-support reviews emphasise that organisational value depends on transparency, process integration and evaluability as much as on isolated accuracy gains [1, 20]. Surveys of post-hoc explanation catalogue attribution methods and their limits [10], while position papers argue for models that are interpretable by design [2, 3, 8]. Applied AI venues such as EAAI likewise host engineering analyses of learning systems and their practical trade-offs [6]. Named channels, declared weights, interaction rules and exportable firing-incidence records instantiate that design stance in a temporal pipeline. What remains comparatively less evaluated is whether those structured levers change interval outcomes when perception is frozen.

Missing data and degraded sensors are routine in machine learning and industrial monitoring [11, 21, 22]. Industrial IoT studies examine classification when sensor readings are absent [13], and multimodal learning has begun to treat severely missing modalities as a first-class training setting [12]. Across these literatures, a recurring systems distinction is often under-specified in video pipelines: a channel may be *unavailable* (α=0) or *available but zero-valued* (c=0, α=1). Conflating the two silently reweights or imputes evidence. The experiments therefore compare explicit availability masking against naive zero-valued evidence under *synthetic* dropout, while disclosing that natural availability on the BoxingVI matrices is $\alpha \equiv 1$. This is an interface evaluation rather than a claim of robustness under natural channel missingness.

Existing work therefore provides powerful learned temporal models, multi-modal systems and rule-oriented decision tools, yet controlled evidence remains scarce on how explicit source availability, simple aggregation choices and interpretable interaction rules behave inside a fixed temporal engineering pipeline when perception is held constant. This paper addresses that gap through same-input ablations, synthetic availability interventions, firing-incidence records and disclosed negative findings, with reproducibility artefacts aligned to community expectations for inspectable ML research [4, 5]. It does not claim a new multi-source aggregation calculus, a general theory of partial observability, an external state-of-the-art detector, or a deployment-ready safety product.

3. System model and pipeline methodology

For each BoxingVI stem (video) and frame t, perception yields a fixed feature row. A shared rule layer maps features to channel confidences $c_{i,t} \in [0, 1]$ and binary availability masks $\alpha_{i,t} \in \{0, 1\}$. Zero-valued evidence is ($c_{i,t}$=0, $\alpha_{i,t}$=1); unavailable evidence is $\alpha_{i,t}$=0 and is excluded from renormalisation. Ground-truth masks derived from annotated punch/impact intervals are stored for evaluation only and are never used to build channels or to tune weights in the checkpoint. Frame-wise aggregation maps the active set $A_t = \{i : \alpha_{i,t}=1\}$ to a score $s_t \in [0, 1]$. Optional interaction rules may add bounded boosts when all required channels jointly exceed thresholds. Scores map to bands $\{\mathrm{LOW}, \mathrm{MEDIUM}, \mathrm{HIGH}, \mathrm{CRITICAL}\}$ with fixed cutoffs. A consolidator merges high-band risk runs into intervals, while a held-fixed strike detector contributes strike intervals independently. The *combined timeline* merges risk and strike intervals under the reference aggregation pipeline (software export identifier: `full_fusion`). A matcher then compares predicted and reference intervals under fixed intersection-over-union (IoU) and temporal tolerance. Aggregation methods change only the risk component; the strike component is identical across methods; reported metrics use the combined

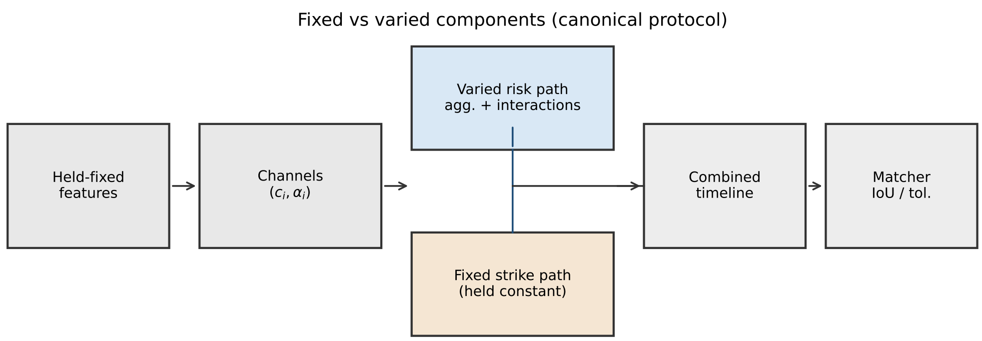

Figure 1: Pipeline with (c, α) channels, fixed strike path and combined timeline.

timeline, so a shared strike mass can attenuate visible aggregation differences (Section 5.4).

Figure 1 summarises fixed versus varied components. Implementation artefacts live in the FightSafe AI research software [23]; this manuscript evaluates a frozen experimental layer over held-fixed feature caches. Supplementary Tables A.8–A.10 list channels, weights, rules and protocol parameters without altering configuration.

On channels with configured weights $w_i > 0$ and $\alpha_{i,t}=1$, the three aggregation schemes are

$$s_t^{\mathrm{eq}} = \frac{1}{|A_t|} \sum_{i \in A_t} c_{i,t}, \tag{1}$$

$$s_t^{\mathrm{w}} = \frac{\sum_{i \in A_t} w_i c_{i,t}}{\sum_{i \in A_t} w_i}, \tag{2}$$

$$s_t^{\max} = \max_{i \in A_t} c_{i,t}. \tag{3}$$

Max is an intentional aggressive stress test that allows one channel to dominate rather than a competitive external baseline. The primary three-way comparison uses interactions enabled. Declarative interaction rules add boosts when all required channels are available and exceed thresholds (Supplementary Table A.9). A *firing incidence* is one frame in which a rule's predicate holds. Rules whose required channels are absent cannot fire; thresholds are not retuned. In the checkpoint, the `surrender_gesture` channel is absent (one rule inactive) and the fall-like inactivity rule records zero firing incidences despite present channels.

Frozen temporal and matching parameters are frames per second (fps) = 30, risk merge gap = 2 frames, minimum event duration $d_{\min}$=0 s, combined-timeline merge = 8 frames, IoU threshold = 0.01 and temporal tolerance = 0.5 s, with evaluation on the combined timeline under risk re-aggregation and a fixed strike path. Permissive IoU/$d_{\min}$ favour recall-oriented matching for short proxy punches; post-hoc stricter matching is reported as sensitivity only (Supplementary Table A.14).

4. Experimental protocol

BoxingVI stems V1–V10 [7] provide punch-interval proxy labels (Table 1); hereafter each stem is treated as one video. Skeleton keypoints are obtained externally and are not redistributed here. Video is the inferential unit (n=10). Reported metrics include pooled micro-precision, micro-recall and micro-F1, video-level macro-precision/recall/F1 (macro-F1), video-level bootstrap 95% intervals (B=1000 resamples of videos), paired mean/median ΔF1, Cliff's δ [24], and paired permutation two-sided p-values [25]. Synthetic frame×channel dropout uses probabilities $p \in \{0.0, 0.1, 0.3, 0.5\}$ with seeds $0, \ldots, 9$ whenever p>0. Power is limited and population generalisation is not claimed. Primary pre-specified contrasts are equal versus weighted versus max, and interactions ON versus OFF; dropout contrasts and leave-one-video-out or matching grids are exploratory or descriptive. Inputs and outputs are checksummed in the checkpoint; a double-run of weighted scoring on V2 is numerically identical; experimental-layer runtime is approximately 81 s after caches are available. Tier A regeneration of tables and figures from frozen canonical CSVs is documented in Section 7.

5. Results

5.1. Aggregation behaviour and video imbalance

Table 2 and Figure 2 compare equal, weighted and max with interactions on. Equal and weighted remain nearly tied on pooled micro-F1, whereas max trades high precision for collapsed recall under single-channel domination of high bands. Paired tests (Table 6) separate weighted from max but not from equal. Figure 3 shows substantial per-video heterogeneity under the shared protocol. These pooled summaries must be read with strong video imbalance:

Table 1: BoxingVI stem statistics (held-fixed matrices).

Video	Frames	Pos. frames	Pos. fraction
V1	1866	1589	0.852
V2	232	130	0.560
V3	813	358	0.440
V4	560	120	0.214
V5	596	253	0.424
V6	46497	10682	0.230
V7	195	109	0.559
V8	199	73	0.367
V9	149	44	0.295
V10	151	30	0.199

Natural availability $\alpha \equiv 1$ on these matrices; missingness is synthetic only.

Table 2: Aggregation on the combined timeline (interactions ON).

Method	Micro-P	Micro-R	Micro-F1	Macro-F1	Macro-F1 95% CI
equal	0.472	0.536	0.502	0.555	[0.471, 0.667]
weighted	0.473	0.535	0.502	0.567	[0.492, 0.674]
max	0.779	0.100	0.178	0.235	[0.134, 0.359]

Max is an aggressive single-channel stress test, not a competitive external baseline.

stem V6 accounts for most pooled ground-truth events and true positives under weighted aggregation (Supplementary Table A.11; Figure 4). Leave-one-video-out recomputation from existing predictions shifts only modestly when V6 is excluded (Supplementary Table A.12), confirming that pooled micro-F1 is not an equal-video average; macro-F1 and paired per-video analyses are therefore emphasised for general behaviour.

5.2. Interaction-rule ablation

Table 3 and Figure 5 compare weighted aggregation with interactions off versus on. Enabling rules increases macro-F1 together with predicted-event counts and HIGH/CRITICAL occupancy. The paired mean ΔF1 is positive with moderate Cliff δ and a borderline permutation p at n=10: protocol evidence, not a general causal claim. Four of six configured rules fire; one cannot activate because `surrender_gesture` is absent; one records zero firing incidences (Table 4).

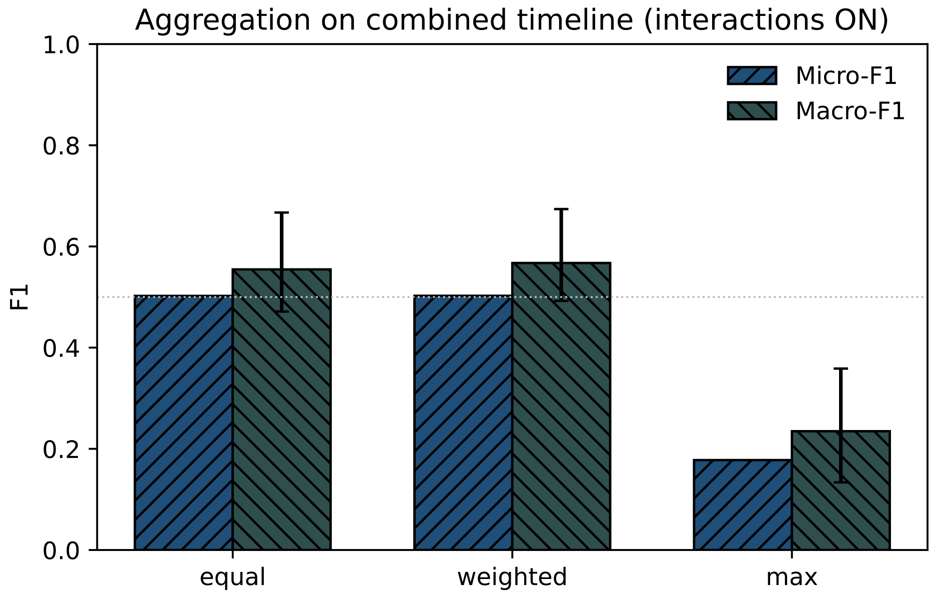

Figure 2: Micro-F1 and macro-F1 for equal, weighted and max (interactions ON).

Table 3: Interaction-rule ablation (weighted aggregation).

Int.	Micro-F1	Macro-F1	95% CI	#pred	H/C%
Off	0.467	0.444	[0.357, 0.520]	75.4	0.01
On	0.502	0.567	[0.492, 0.674]	107.1	12.36

H/C%: mean HIGH/CRITICAL frame share. Paired $p \approx 0.048$ (n=10).

5.3. Synthetic availability masking

Table 5 and Figure 6 report synthetic frame×channel dropout. Natural availability is $\alpha \equiv 1$; missingness is induced. At p=0.5, explicit availability masking remains near the no-dropout baseline and is less degraded than naive zero-valued evidence. Descriptive diagnostics (Section 6) associate the small explicit increase with higher pooled recall and higher false-positive mass, concentrated on V6; no unique causal mechanism is established from the frozen artefacts.

5.4. Timeline coupling and sensitivity analyses

Using frozen prediction exports, strike-only and risk-only timelines evaluate below the combined timeline under the canonical matcher (Supplementary

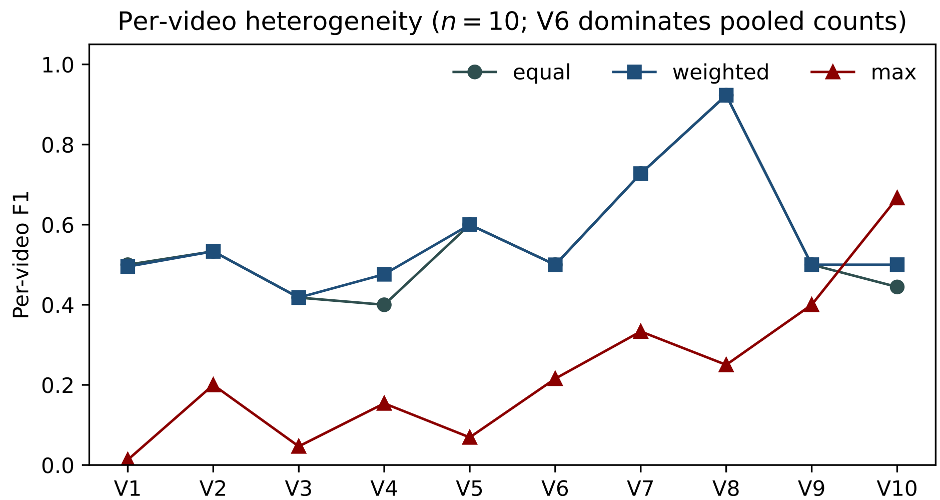

Figure 3: Per-video F1 under the shared protocol (n=10).

Table 4: Interaction-rule firing-incidence totals (interactions ON).

Rule	Total firing incidences
limb+instability	15180
low_guard+strike	8881
low_guard+instability	6163
facing_away+instability	4105
surrender_like (inactive)	0
fall_like+inactivity	0

Frame-level firing incidences; `surrender_gesture` channel absent.

Table A.13). The fixed strike component therefore carries substantial matched mass and can dilute risk-only aggregation differences on the reported metric. Exported failure types are dominated by false negatives and false positives, with additional fragmentation and merged-interval errors (Table 7; Figure 7). Max suppresses many positives (high FN) with few predictions, so raw counts are not comparable across methods without denominators. Re-scoring frozen weighted combined-timeline predictions under stricter IoU or zero temporal tolerance reduces pooled micro-F1 (Supplementary Table A.14). These matching analyses are post-hoc and do not replace the canonical protocol.

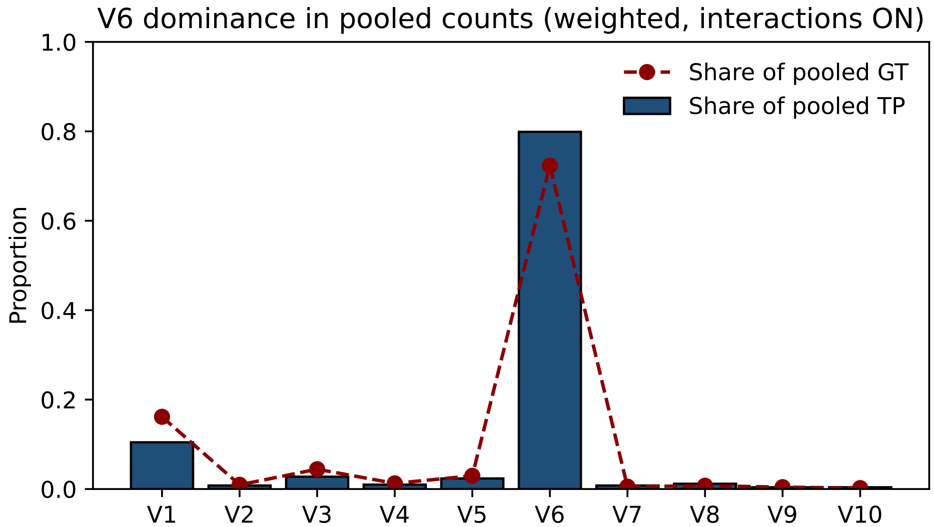

Figure 4: Share of pooled TP and GT events by video (weighted, interactions ON).

6. Discussion and limitations

The near-tie between equal and weighted aggregation implies that, once channels and the consolidator are fixed, fine weight tuning is lower priority than validating interaction logic, availability contracts, consolidator settings or label definitions—especially when one video dominates pooled mass and a fixed strike component dilutes risk-only contrasts on the combined timeline. Max is retained only as a stress test: single-channel domination saturates high bands and collapses recall, so it is not a production default or an external state-of-the-art baseline. Absence of velocity-threshold or independent deep detectors under matching-provenance constraints is intentional honesty rather than omission of available comparable runs. Interaction enablement is best read as a protocol-limited lever with exportable firing-incidence records: useful for inspectability, but not a multiplicity-corrected causal claim at n=10.

Industrial sensing and multimodal learning treat missing measurements as routine rather than exceptional [11, 12, 21]. The practical implication is to keep an explicit α interface even when natural matrices are fully observed ($\alpha \equiv 1$). Under synthetic dropout, naive zero-valued encoding is the more fragile contract; the modest explicit uptick relative to the no-dropout baseline is descriptive only and coincides with higher recall and false-positive mass rather than a proven robustness mechanism. Candidate explanations—

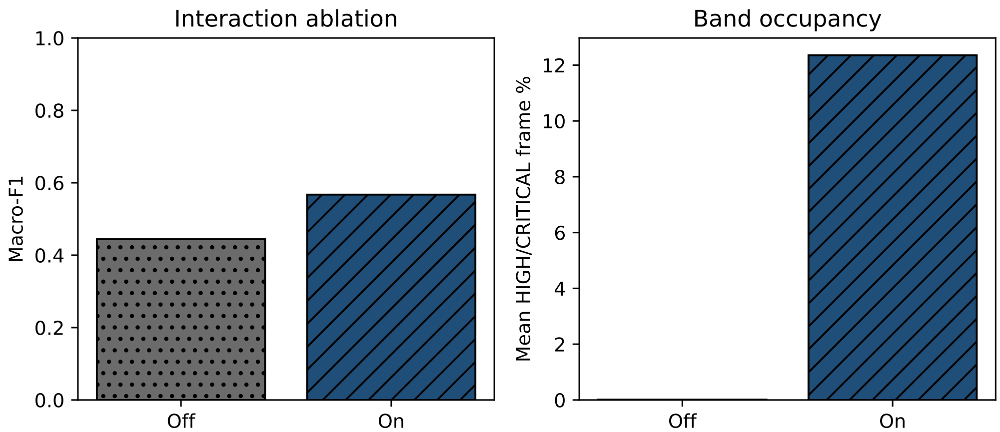

Figure 5: Interaction ablation: macro-F1 and mean HIGH/CRITICAL share.

Table 5: Synthetic dropout under weighted aggregation.

p	Encoding	Micro-F1	SD	Micro-P	Micro-R
0.0	No dropout	0.502	—	0.473	0.535
0.1	Explicit masking	0.506	0.003	0.479	0.536
0.1	Naive zero	0.499	0.004	0.479	0.520
0.3	Explicit masking	0.507	0.006	0.477	0.542
0.3	Naive zero	0.483	0.005	0.499	0.468
0.5	Explicit masking	0.509	0.007	0.461	0.569
0.5	Naive zero	0.471	0.003	0.523	0.428

Means over seeds for $p>0$; natural $\alpha\equiv 1$.

renormalisation over fewer active channels, altered merges with the fixed strike path, and permissive matching—remain unresolved in the frozen artefacts, so Figure 6 must not be read as graceful degradation under natural channel missingness. Near-ties, brittle baselines and unresolved mechanisms are retained because they reduce wasted tuning and clarify which pipeline stages deserve instrumentation [4, 5, 8].

Several limits follow directly from the frozen protocol. The sample comprises $n=10$ videos, so statistical power is limited and the interaction $p=0.048$ is borderline and multiplicity-unadjusted. Pooled micro-F1 is not an equal-video summary under V6 dominance. The combined timeline includes a fixed strike component that can attenuate risk-aggregation contrasts. Labels are

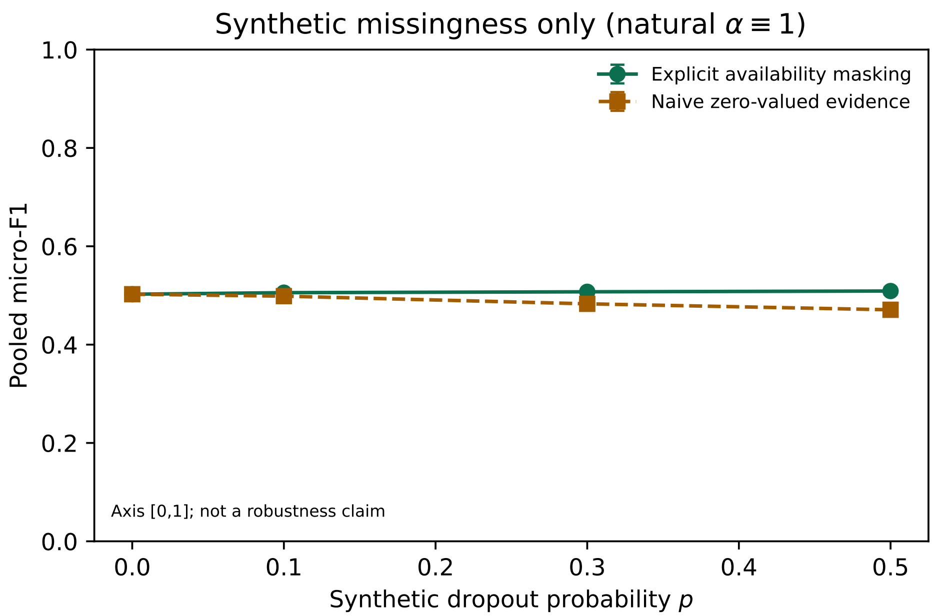

Figure 6: Pooled micro-F1 versus synthetic dropout probability (natural $\alpha \equiv 1$).

punch/impact proxies rather than clinical or safety ground truth. Natural availability is $\alpha \equiv 1$; missingness results are synthetic; and the small explicit micro-F1 increase is descriptive only. Weights, thresholds and rules are hand-authored, so historical tuning leakage cannot be ruled out. The matcher is permissive (IoU threshold 0.01, $d_{\min}=0$) and stricter matching lowers F1 in post-hoc checks. There is no independent external dataset, operator study or deployment claim; one interaction rule cannot fire and another records zero firing incidences; and Tier B full re-execution requires redistributable derived feature caches outside the public Tier A artefact (Section 7).

7. Conclusion

An availability-aware, interpretable temporal event pipeline was evaluated on BoxingVI as an engineering case study with disclosed limits. Equal and weighted aggregation behave similarly; max is a brittle stress test; interaction rules show a protocol-limited effect; explicit availability masking is less fragile than naive zero-valued evidence under synthetic dropout only; and pooled metrics are dominated by one stem and shaped by a fixed strike component.

Table 6: Paired video-level comparisons.

Comparison	n	Mean ΔF1	Cliff δ	p
Weighted − equal	10	0.013	0.00	0.489
Weighted − max	10	0.332	0.80	0.007
Interactions ON − OFF	10	0.123	0.40	0.048
Explicit − naive (p=0.1)	10	0.029	0.60	0.017
Explicit − naive (p=0.3)	10	0.069	0.60	0.073
Explicit − naive (p=0.5)	10	0.128	0.60	0.032

Permutation two-sided p; Cliff's δ. Dropout contrasts exploratory.

Table 7: Failure taxonomy counts (shared matcher).

Method	Approx.#pred	FP	FN	Fragment.	Merged
equal	1076	568	439	36	142
weighted	1071	564	440	35	141
max	122	27	852	15	85
weighted (int. off)	754	357	550	14	95

Counts are not normalised across methods because prediction cardinality differs (especially max).

These findings support an EAAI-oriented narrative in which inspectable contracts, controlled ablations and published negative results are central engineering contributions.

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Institutional affiliations are as listed on the title page. This manuscript is not a medical-device evaluation.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used generative artificial intelligence tools and AI-assisted editorial tools solely to support manuscript writing and editorial quality control. Assistance was limited to language improvement, grammar and style refinement, restructuring of sections for clarity, readability edits, consistency checking, bibliography consistency checks, literature exploration for Related Work positioning, checking of bibliographic and

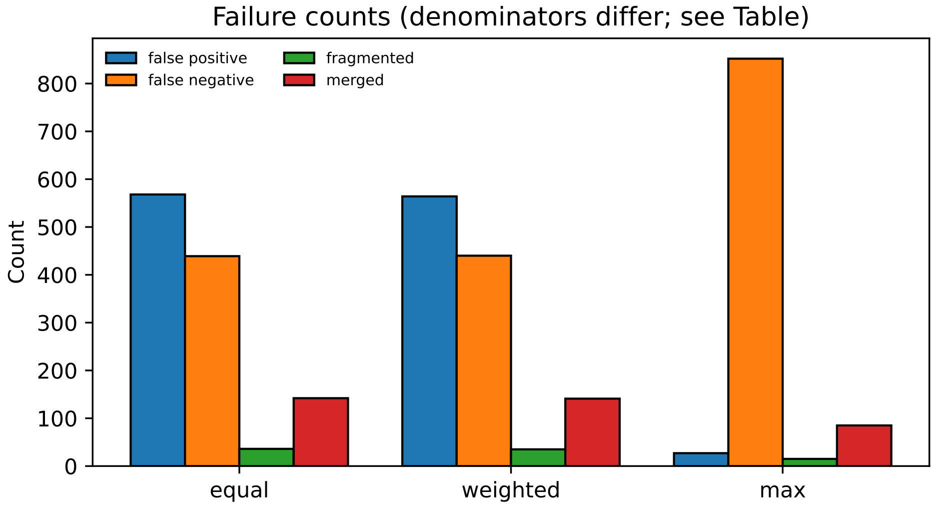

Figure 7: Failure taxonomy counts for equal, weighted and max.

DOI metadata, preparation of reproducibility documentation, and editorial review of manuscript text. Generative AI was not used to generate experimental data, execute experiments, modify canonical numerical results, produce figures or tables without subsequent author verification, make scientific decisions, derive statistical conclusions, or replace scientific judgement. All experiments were designed and executed by the authors; all software implementations were produced and validated by the authors; and all reported numerical results were checked independently against the frozen canonical checkpoint. All scientific interpretations and conclusions are the sole responsibility of the authors. The authors reviewed and validated every AI-assisted contribution before its inclusion in the manuscript and take full responsibility for the content of the article.

Data and code availability

BoxingVI punch-interval annotations used for evaluation may be redistributed with the software under the data notice; skeleton keypoints must be obtained from the dataset authors [7]. Experimental aggregates are archived under the canonical identifier `run_20260730_005150`. Software context: [23]. The public GitHub/Zenodo artefact supports regeneration and verification of the reported tables and figures from canonical result files (Tier A). Figures and tables are regenerated with `scripts/generate_eaai_assets.py` using repository-relative paths; configs, aggregation implementations, environment freeze and SHA-256 checksums are included. Tier B inputs (`features_cache`) are not redistributed with the public artefact and require separate upstream or institutional clearance. A software archive accompanying this manuscript is deposited at DOI `10.5281/zenodo.21698326`; this manuscript does not claim end-to-end feature reconstruction from raw media.

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Table A.8: Channel inventory and fixed weights.

Channel	Weight	Related rules
fast_downward_motion	0.0828	—
large_torso_angle	0.0828	—
prolonged_low_posture	0.0828	—
high_instability	0.0828	low_guard+instability, facing_away+instability, limb+instability
post_fall_low_movement	0.0828	fall_like+inactivity
low_guard	0.0736	low_guard+instability, low_guard+strike
facing_away	0.0736	facing_away+instability
reaction_delay_after_ impact	0.0736	—
loss_of_control	0.0860	—
clear_danger_fall	0.0860	fall_like+inactivity
intervention_urgent	0.0952	—
high_risk_guard_strike	0.0600	low_guard+strike
limb_anomaly	0.0800	limb+instability
surrender_gesture	0.0600	surrender_like

Pose/feature heuristics; natural $\alpha \equiv 1$ on checkpoint matrices.

Appendix A. Supplementary configuration and sensitivity tables

Table A.9: Configured interaction rules.

Rule	Channels	Boost	Thr.	Status
low_guard+instability	low_guard, high_ instability	0.06	0.08	fires
facing_away+instability	facing_away, high_ instability	0.07	0.08	fires
low_guard+strike	low_guard, high_risk_ guard_strike	0.07	0.08	fires
fall_like+inactivity	clear_danger_fall, post_ fall_low_movement	0.10	0.08	0 incidences
surrender_like	surrender_gesture	0.12	0.25	inactive
limb+instability	limb_anomaly, high_ instability	0.08	0.08	fires

Table A.10: Fixed temporal and matching parameters.

Parameter	Value
FPS	30
Risk merge gap (frames)	2
Risk $d_{\min}$ (s)	0
Combined merge (frames)	8
Matcher IoU threshold	0.01
Matcher temporal tolerance (s)	0.5
Band mins (M/H/C)	0.25 / 0.5 / 0.75
Evaluation timeline	Combined
Strike component	Fixed baseline cache
Bootstrap B	1000 (video-level)

Table A.11: Per-video contribution (weighted, interactions ON).

Video	GT	TP	FP	FN	Share GT	Share TP	F1
V1	153	53	8	100	0.162	0.105	0.495
V2	9	4	2	5	0.010	0.008	0.533
V3	42	14	11	28	0.044	0.028	0.418
V4	12	5	4	7	0.013	0.010	0.476
V5	28	12	0	16	0.030	0.024	0.600
V6	685	405	531	280	0.723	0.799	0.500
V7	5	4	2	1	0.005	0.008	0.727
V8	7	6	0	1	0.007	0.012	0.923
V9	4	2	2	2	0.004	0.004	0.500
V10	2	2	4	0	0.002	0.004	0.500

Table A.12: Leave-one-video-out pooled micro-F1.

Excluded	Pooled micro-F1	TP	FP	FN
V1	0.503	454	556	340
V2	0.502	503	562	435
V3	0.505	493	553	412
V4	0.503	502	560	433
V5	0.501	495	564	424
V6	0.514	102	33	160
V7	0.501	503	562	439
V8	0.500	501	564	439
V9	0.502	505	562	438
V10	0.502	505	560	440

Descriptive sensitivity only; not a new experimental run.

Table A.13: Timeline components (weighted, interactions ON).

Timeline	Micro-F1	P	R	TP	FP	FN	#pred
Combined	0.502	0.473	0.535	507	564	440	1071
Risk-only	0.410	0.354	0.488	462	844	485	1306
Strike-only	0.466	0.533	0.415	393	345	554	738

Strike path is held fixed across aggregation methods; risk path is re-aggregated.

Table A.14: Post-hoc matching sensitivity.

IoU	Tolerance (s)	Micro-F1	P	R	TP	FP	FN
0.01	0.5	0.502	0.473	0.535	507	564	440
0.10	0.5	0.485	0.457	0.516	489	582	458
0.30	0.5	0.409	0.386	0.436	413	658	534
0.50	0.5	0.277	0.261	0.295	279	792	668
0.01	0.0	0.384	0.361	0.409	387	684	560
0.10	0.0	0.295	0.278	0.315	298	773	649
0.30	0.0	0.159	0.149	0.169	160	911	787

Does not replace the canonical protocol (IoU=0.01, tolerance=0.5 s).